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Evaluation of 4H-SiC MOSFET transfer characteristics using machine-learning techniques

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The performance of 4H silicon carbide (SiC) MOSFETs critically depends on the quality of the SiC/silicon oxide interface, which typically contains a high density of interface traps. To solve this problem, fast and reliable characterization methods are required. The commonly used evaluation schemes for 3-terminal transfer characteristics, however, neglect the presence of interface traps. Here, a method based on machine-learning techniques is presented which extracts reliable performance parameters from transfer characteristics of 4H-SiC MOSFETs including a quantitative estimate of the density of interface traps. This method is successfully validated by comparison with Hall-effect measurements and applied to various MOSFET types.

Introduction: Silicon carbide (SiC) MOSFETs have been shown to provide a significant advantage compared to silicon-based insulated gate bipolar transistors (IGBTs) [1]. However, a prevalent problem in 4H-SiC MOSFETs is a high density of defects at the semiconductor-oxide interface resulting in a high density of states particularly near the conduction band [2, 3]. Post-oxidation annealing (POA) in nitric oxide (NO) is often used to reduce the density of interface traps (D_{it}) [4, 5]. Further methods to reduce the density of interface traps D_{it} as well as methods to measure them are still being developed [6, 7]. Hence, a quick characterization of interface traps is of highest value. A usual way to evaluate the performance of a MOSFET is to extract the threshold voltage and charge carrier mobility from 3-terminal characteristics [8, 9]. The commonly used evaluation methods estimate the density of inversion charge carriers via a plate capacitor approximation and neglect the D_{it}. Thus, the number of charge carriers is overestimated and their mobility is underestimated [10, 11]. To tackle these problems Hauck et al. demonstrated a D_{it}-respecting method for the evaluation of 3-terminal transfer characteristics [12]. In that method, a model for the specific D_{it} distribution of 4H-SiC is used for the calculation of available charge carriers. Using this method, however, includes extensive numerical calculations. Furthermore, when fitting the model to measurement data, one needs to find good starting values by repeatedly simulating transfer characteristics. The method turned out to be predictive when comparing to Hall-effect results, being the gold standard method for such measurements, and it can also be applied to difficult geometries as in the case of trench MOSFETs where Hall-effect measurements cannot be conducted.

In this manuscript, we introduce machine-learning techniques for the evaluation of transfer characteristics, based on the D_{it}-respecting method. We aim to reduce computation time and eliminate the need for finding starting values for a fit, thus providing a fast, reliable and easy way for the evaluation of device performance parameters and an estimation for the density of interface traps from transfer characteristics in the peculiar and technologically important case of 4H-SiC MOSFETs.

Methods: Our method of choice is a random forest regressor implemented via the python-package scikit-learn [13–15]. For the training of the regressor, we simulated transfer characteristics following the calculation principles of the D_{it}-respecting method with varying parameters. For the calculations we used the channel geometry, i.e. width and length, of the respective MOSFETs. The varying parameters were the channel mobility μ, the series resistance R_{S}, the total fixed oxide charge Q_{ox}, and the parameters D_0, D_1, σ and E_0 of the D_{it} distribution [12]:

$$D_{\mathrm{it}}(E) = D_0 + \frac{D_1}{\sigma\sqrt{\pi/2}} \cdot \exp\left(-2 \cdot \frac{(E-E_0)^2}{\sigma^2}\right). \quad (1)$$

Since the mobility, compared to the number of charge carriers, is very weakly dependent on the gate voltage, we used a constant mobility.

Table 1. Variation range of the simulation parameters for the training data.

Parameter	Trench	NO-POA Hall	O_2-POA Hall
R_{S} (Ω)	0.002–0.12	0–1	0–1
μ ($\mathrm{cm^2\,V^{-1}\,s^{-1}}$)	30–100	1–32	1–20
Q_{ox} (10^{-7} As $\mathrm{cm^{-2}}$)	0.9–1.5	0–4	−4–0
D_0 (10^{12} $\mathrm{cm^{-2}\,eV^{-1}}$)	0.7–5	0.01–8	3–70
D_1 (10^{12} $\mathrm{cm^{-2}}$)	3.5–5.5	0.01–90	30–90
E_0 (eV)	0.01–0.15	0–0.12	0–0.2
σ (eV)	0.1–0.2	0.18–0.35	0.1–0.35

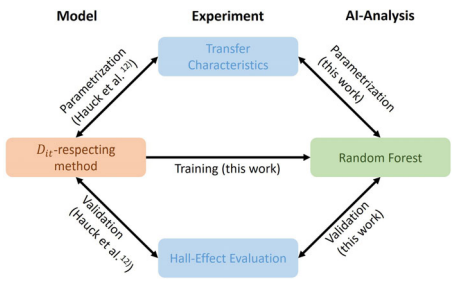

Fig. 1 *Schematics of the procedure of this work. Hauck et al. devised the D_{it}-respecting method to parameterize transfer characteristics with it and validated it with Hall-effect measurements. In this work, we use the D_{it}-respecting method to train a random forest, which is again used to parameterize transfer characteristics and is validated by comparison with Hall-effect measurements*

Hereby we also avoid the assumption of prominent scattering mechanisms in the channel. Furthermore, a single value can be easily compared to the other three-terminal methods. In general it is possible to use any model, but this would lead to more assumptions and more parameters, the number of which should be kept as small as possible for a proof of concept.

The obtained transfer characteristics were used to train the random forest with the voltage-dependent drain current and its logarithm as input parameters. The random forest should then yield the parameters used for the simulation of the respective transfer characteristics. A rough grid search using simulated validation data was performed to optimize the hyperparameters of the random forest, i.e. number of trees, forest depth and feature number per split [15]. Subsequent to the training, the random forest is put to test with measurement data of real devices. The parameters yielded by the random forest are then used to simulate transfer characteristics, which are compared with the measurement data.

We investigated three different types of devices using quasi-static current-voltage measurements within the linear regime of the devices' output characteristics: a high-mobility trench MOSFET test structure and two MOSFETs in a Hall-bar design with POA in NO-atmosphere and O_2-atmosphere respectively. The usage of the MOSFETs in Hall-bar design allows for a good assessment of the D_{it} distribution and the mobility via Hall-effect measurements [16]. For each type of MOSFETs, we randomly varied the parameters in a physically reasonable range as listed in Table 1. To further examine the performance of the random forest, the results of the machine learning evaluation are then re-used as starting parameters for a least squares fit. Fig. 1 shows the schematics of the overall procedure, including the work of Hauck et al.

Results: The transfer characteristics obtained from the results of the random forest evaluation and the subsequent numerical fit are depicted in Fig. 2 together with the measurement data. The random forest results agree already quite well with the measured data in both, linear and logarithmic scale. The subsequent numerical fit using the random forest results as start parameters resembles the measurement more closely, in particular for the Hall-bar MOSFETs. A comparison of the parameters obtained by random forest and the subsequent least squares fit is given in Table 2.

Compared to the trench MOSFET test structure, the typical drain currents of the Hall-MOSFET structures are substantially lower. This

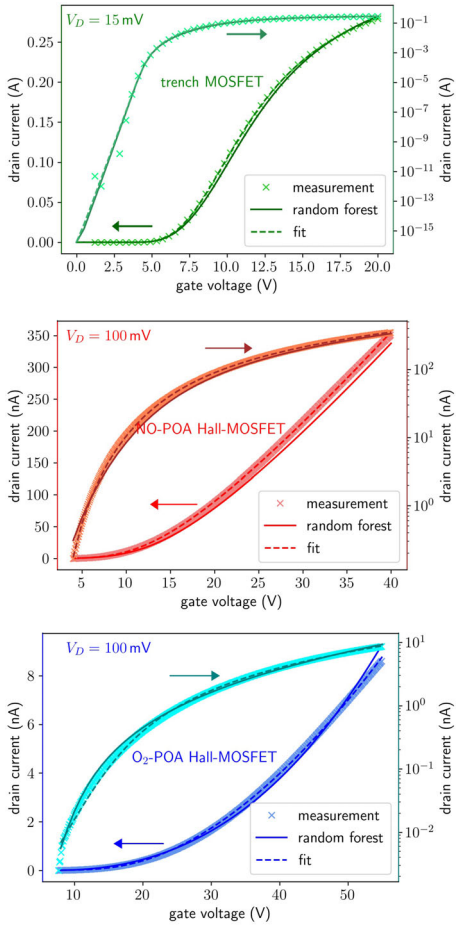

Fig. 2 *Transfer characteristics, measurement with drain voltage in the linear regime of the output characteristics, simulation with parameters yielded by the random forest regressor and by a consecutive fit for a trench MOSFET test structure, a Hall-bar MOSFET with POA in NO and a Hall-bar MOSFET with POA in O_2. The consecutive fit resembles the measurement rather closely for all investigated MOSFETs. The curves obtained from the random forest are a bit more off, yet provide a good estimate in both, the linear and the logarithmic scale*

Table 2. Parameters obtained by the random forest regressor (rf) and the consecutive numerical fit

	Trench		NO-POA Hall		O_2-POA Hall	
Parameter	rf	fit	rf	fit	rf	fit
R_S (Ω)	0.040	0.041	0.44	0.99	0.53	0.53
μ (cm^2 V^{-1} s^{-1})	56	58	28	29	12	10
Q_{ox} (10^{-7} As cm^{-2})	1.4	1.4	2.9	2.2	-0.3	0
D_0 (10^{12} cm^{-2} eV^{-1})	1.1	1.2	1.7	1.1	12	26
D_1 (10^{12} cm^{-2})	4.5	4.8	11	11	59	88
E_0 (meV)	33	42	64	74	58	81
σ (eV)	0.16	0.17	0.25	0.22	0.15	0.12

results from the channel geometry required for Hall-effect measurements. Further, the different POAs affect the drain current strongly, which results mainly from the drastically different D_{it} leading to charge carrier trapping.

The mobility from random forest and fit can further be compared to classical three-terminal methods, like the field effect mobility and the method by Ghibaudo [8, 9]. An overview together with the Hall-mobility at maximum gate voltage is given in Fig. 3. It becomes apparent, that the two classical 3-terminal evaluation schemes yield smaller values across the board than Hall effect or D_{it}-respecting method, particularly for the device with POA in O_2 atmosphere.

An advantage of the D_{it}-respecting method is that it also gives an estimation for the density of interface traps distribution D_{it} in the bandgap.

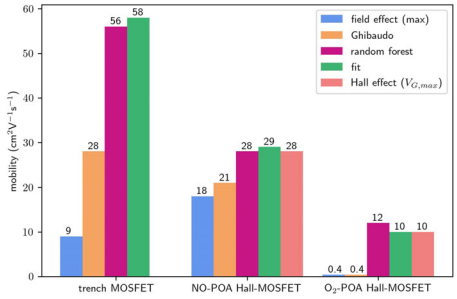

Fig. 3 *Charge carrier mobility values of the different evaluation schemes in comparison. For the Hall effect mobility, the value at the highest gate voltage was used, i.e. $V_G = 40$ V for the NO-POA MOSFET and $V_G = 55$ V for the O_2-POA MOSFET, whereas the maximum value was used for the field effect mobility. For all MOSFETs, the field effect mobility is lowest, followed by the Ghibaudo method, while the values are quite similar for random forest, fit and Hall effect measurements. Especially for the MOSFET with O_2-POA it becomes clear that the classical three-terminal methods are far off, yielding absurdly low values*

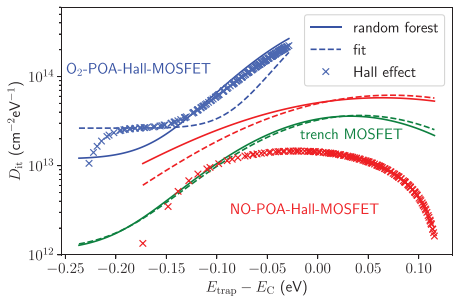

Fig. 4 *Density of interface states as a function of energy in the bandgap of 4H-SiC obtained for all three MOSFETs by the random forest regressor, a consecutive numerical fit and from Hall-effect measurements. The trench MOSFET test structure, which naturally is not accessible by Hall-effect investigation, is depicted in green, the NO-POA Hall-MOSFET in red and the O_2-POA Hall-MOSFET in blue. Fit and random forest deliver similar D_{it} for all MOSFETs. As can be expected, the trench MOSFET test structure has the lowest D_{it}, followed by the one with POA in NO. The D_{it} of the NO-POA MOSFET is slightly overestimated, whereas the D_{it} of the O_2-POA MOSFET is described very well by the random forest and slightly underestimated by the numerical fit*

The D_{it} resulting from the random forest and from the numerical fit are shown in Fig. 4 together with the D_{it} obtained from Hall-effect measurements. The random forest D_{it} of the O_2-POA MOSFET agrees very well with the Hall data, while the D_{it} distribution from the numerical fit is a little too low. For the NO-POA Hall-MOSFET the random forest as well as the numerical fit result in a little too high D_{it} compared to the Hall-effect data. The deviation results from the comparatively low D_{it}, which has only a minor impact on the transfer characteristics. Consequently, the determination of D_{it} from the transfer characteristics is subject to a larger uncertainty. For the trench MOSFET test structure, for which Hall-effect investigation is not possible, the two D_{it} distributions resulting from the random forest and the subsequent fit match quite well.

Using varying drain voltages within the linear regime of output characteristics, the same random forest still delivers very good results. For this, the measured drain current is simply rescaled as $I'_D = V_{D,train}/V_{D,meas} \cdot I_{D,meas}$. The resulting parameters differ from the original values up to 3% mostly however less than 1%.

Discussion: In general, the parameters obtained by the random forest regressor and from the consecutive least squares fits are quite similar. This demonstrates that the random forest provides a valuable first guess, which is again emphasized by the good match of data and simulated characteristics using the obtained parameters.

The tremendous difference regarding the mobility values between random forest/fit and classical 3-terminal methods, as seen in Fig. 3, shows the value of the new method. With the introduction of the D_{it}

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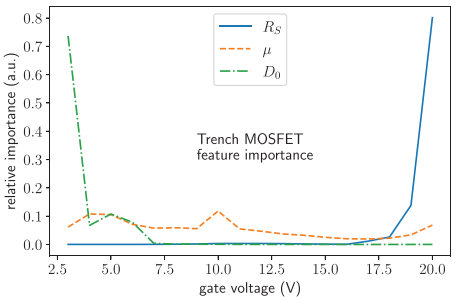

Fig. 5 *Feature importance of the random forest for the trench MOSFET test structure, exemplarily depicted for the series resistance R_S, the mobility μ and the constant part of the density of interface traps D_0. The random forest employs mostly the current values at high gate voltages to determine the series resistance, while using the values at low gate voltages for the constant part of D_{it}. The charge carrier mobility, however, depends on all current values almost equally*

into the calculations, the density of charge carriers is not overestimated as before and thus a better estimation of the mobility is possible. Furthermore, the D_{it} distribution allows for a better assessment of possible reasons for device deficiencies.

The advantage of the random forest method becomes especially clear when looking at computation time: the evaluation of a transfer characteristic takes less than a second with the random forest regressor. For comparison, a numerical least squares fit takes at least 10 min up to several hours. Moreover, for the machine learning approach, no starting values are required reducing working time even further by eliminating the need for tuning the starting parameters by hand.

Apart from that, a machine learning approach provides even further ways of analyzing the data. As an example of one such possibility, Fig. 5 shows the analysis of so-called feature importance, i.e. how much one feature, in our case a current value at some voltage, contributes to the estimation of the return parameter. As one might expect, the series resistance R_S depends mainly on the values at higher gate voltages where significant electrical currents lead to a voltage drop across R_S. The constant part D_0 of the D_{it} distribution on the other hand dominates at the lower gate voltages, namely the sub-threshold regime. Lastly, the mobility depends similarly on the values at all voltages. These extended capabilities may provide further insight into physical concepts.

Conclusion: In conclusion, we used a random forest regressor as a machine-learning approach for the evaluation of transfer characteristics of 4H-SiC MOSFETs. For the training of the random forest, we simulated transfer characteristics using the D_{it}-respecting method of Hauck et al. We compared the random forest evaluation with a consecutive least squares fit and with classical 3-terminal methods on three different device types. In addition, we compared the results to parameters obtained from Hall-effect measurements, which give the most reliable results. The random forest performs quite well in reproducing the transfer characteristics and in the estimation of charge carrier mobility and D_{it}. While the results of the least-squares fit may be more exact, the random-forest method eliminates the necessity for well-chosen starting parameters and reduces the computation time significantly. Moreover, we have discussed further capabilities of a machine learning approach by analyzing feature importance providing further insights into the underlying physics.

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